esprit

Se former autrement

DATA ENGINEERING PROJECT

Class: **3IA**

cdio

EUR-ACE® Délivrée par la Commission des Titres d'Ingénieur

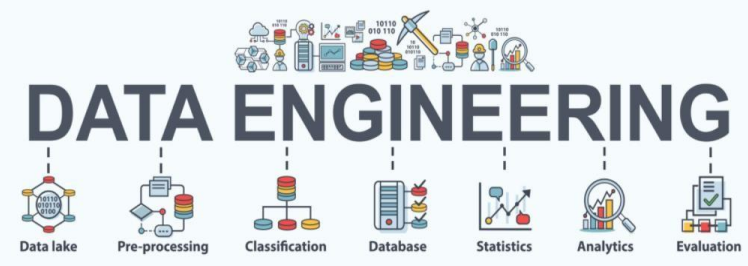
CONFÉRENCE DES GRANDES ÉCOLES

UNESCO Chair "Project-based learning" ESPRIT School of Engineering, Tunisia

Table of contents

I. About the project
II. Major Requirements
III. Project goals
IV.Overview
V.TO DO

About the project



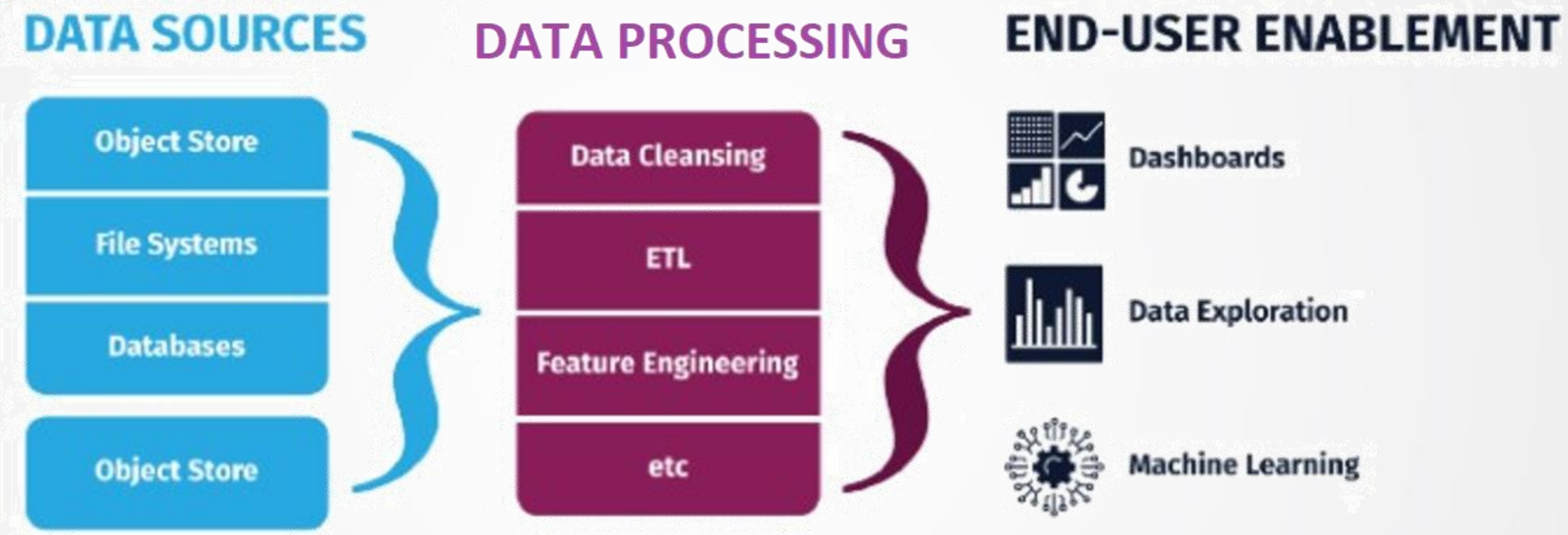


"Data engineering is a multidisciplinary field with applications in control, decision theory, and bioinformatics. It is a new discipline that focuses on the design, implementation, and maintenance of distributed information systems." [Eze 2018]

"Data engineering is systems-oriented and it is a young field. The relevant technologies include communications systems, security, database systems, and distributed systems." [Frieder and Shuey 2015]

"Data engineering's objective is meeting the needs of information systems. It takes into account the interaction among data actions." [Chun 2012]

About the project



Phases of Data Engineering [medium 2019]

Project Goals

1. *Subject* matter *expertise*;
2. Conduct 'data project' from ***Exploration → Production***;
3. *Design, build and understand infrastructures and systems* responsible for collecting, storing, analyzing and interpreting various data in a relevant way;
4. Give the business the necessary tools to take the most appropriate *strategic and operational decisions*, using data analysis as the basis.

Without data engineering, it would be tough to make sense of massive data amounts. It is a crucial aspect of a company's growth and anticipating future trends.



Project Goals

This project will allow you to acquire skills in data engineering, data science, and data analysis

Data Scientist



uses statistics and machine learning to make predictions and answer key business questions

Data Engineer



build and optimize the systems that allow data scientists and analysts to perform their work

Data Analyst

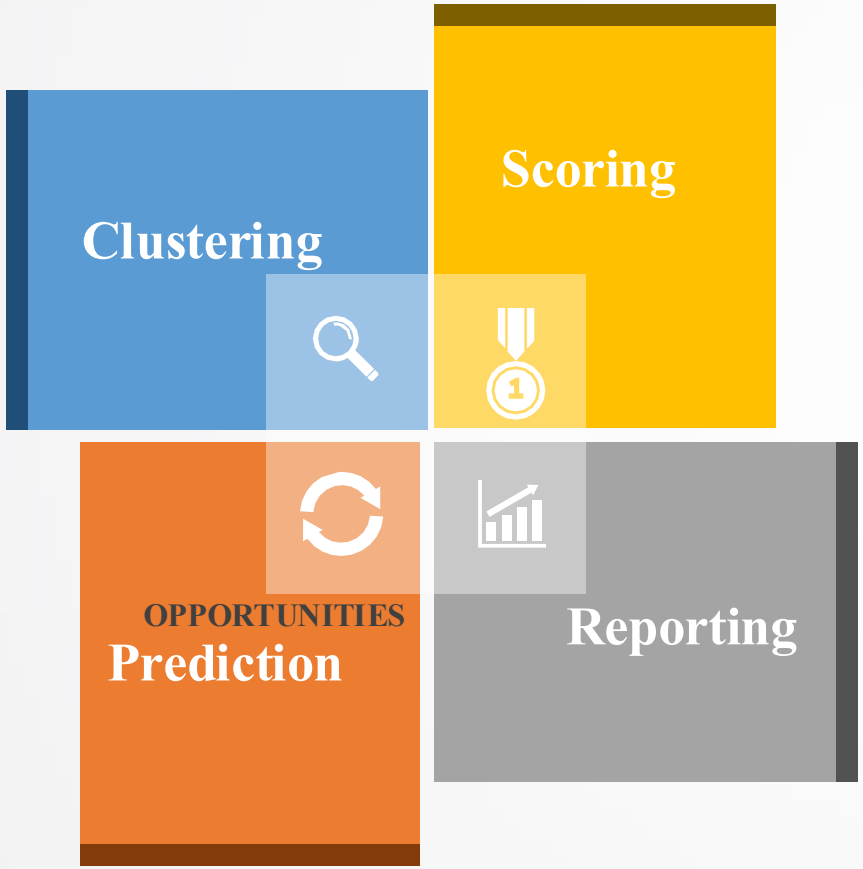


deliver value by taking data, communicating the results to help make business decisions



Business objectives & Data understanding

Business Objectives



Clustering

- Determining customer profiles by similarity and by their financial behavior

Scoring

- Determining customer sensitivity to interactions with the bank
- Determining customer's sensitivity to social networks
- Determining customer sensitivity to the general press

Prediction

- Prediction of business opportunity
- Prediction of commercial risk

Reporting

- Restitution in the form of dashboards of the various elements obtained from the data warehouse and the Clustering and the Prediction











Data understanding



Customer Activity

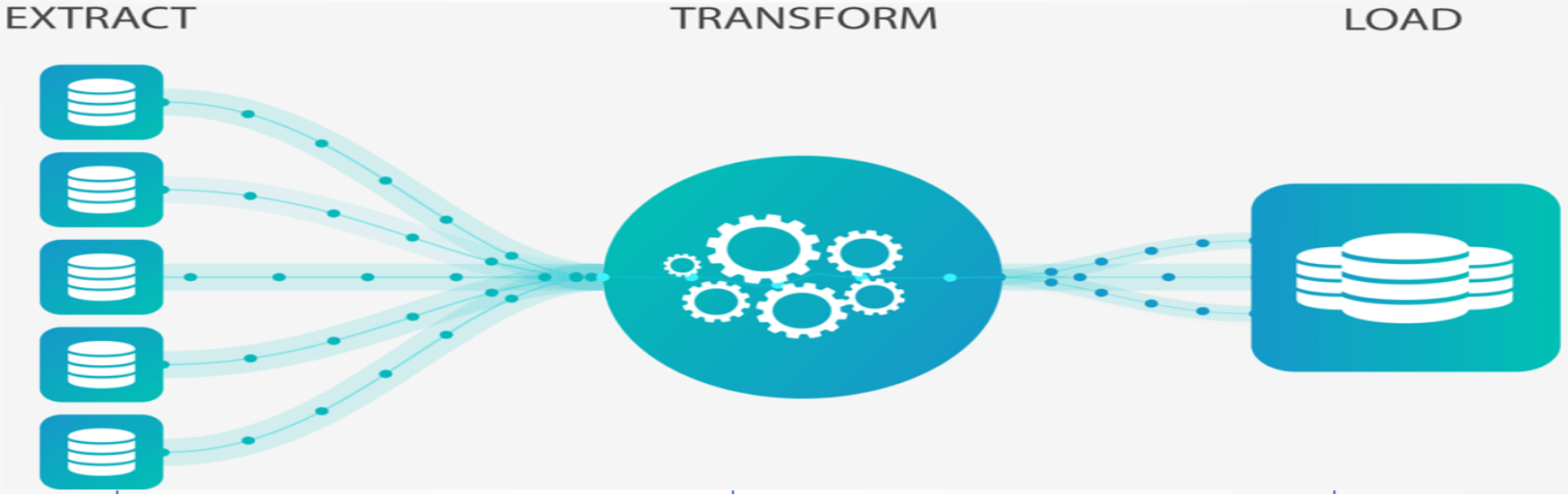
Referential **Clients**

Financial **Transactions**



Data collection & Data preparation

ETL process

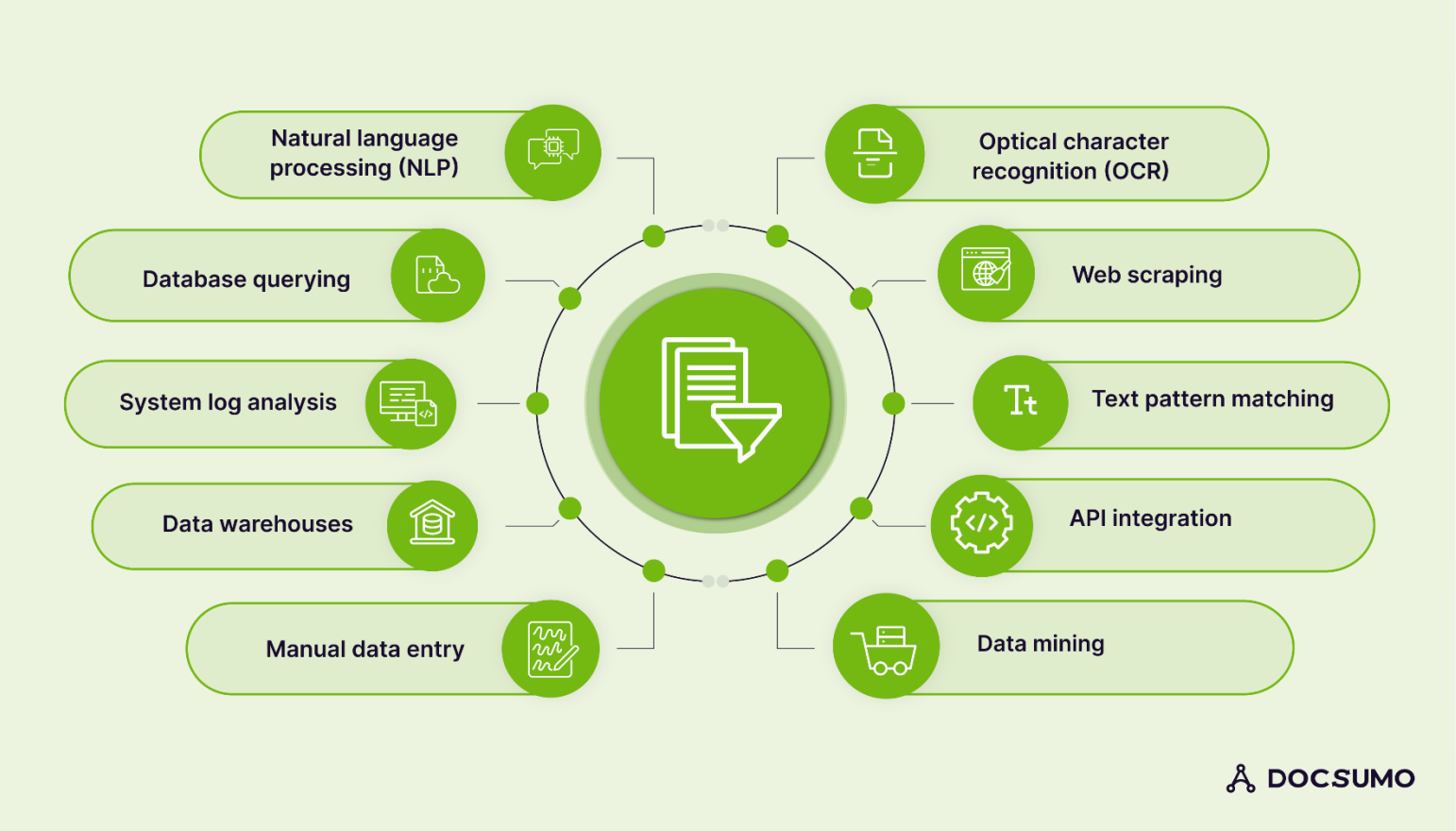


Extracting data from source systems such as databases, files, or APIs. The data is collected in its raw form for further processing

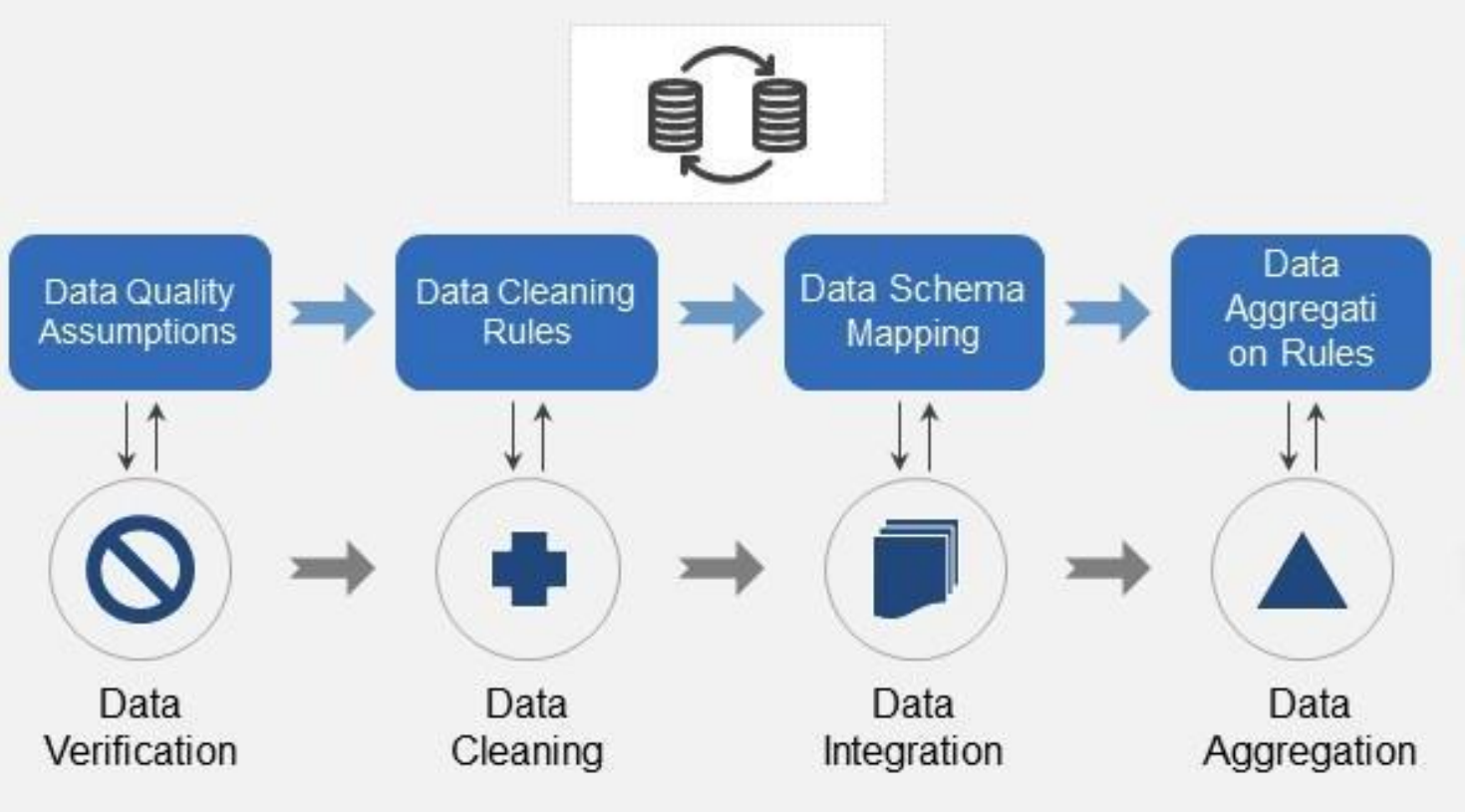
The data extracted is raw and not usable in its original form.Therefore it needs to be cleansed, mapped and transformed. Transformations may include filtering, aggregating, joining, or converting data types.

Loading data into the target data warehouse is the last step of the ETL process

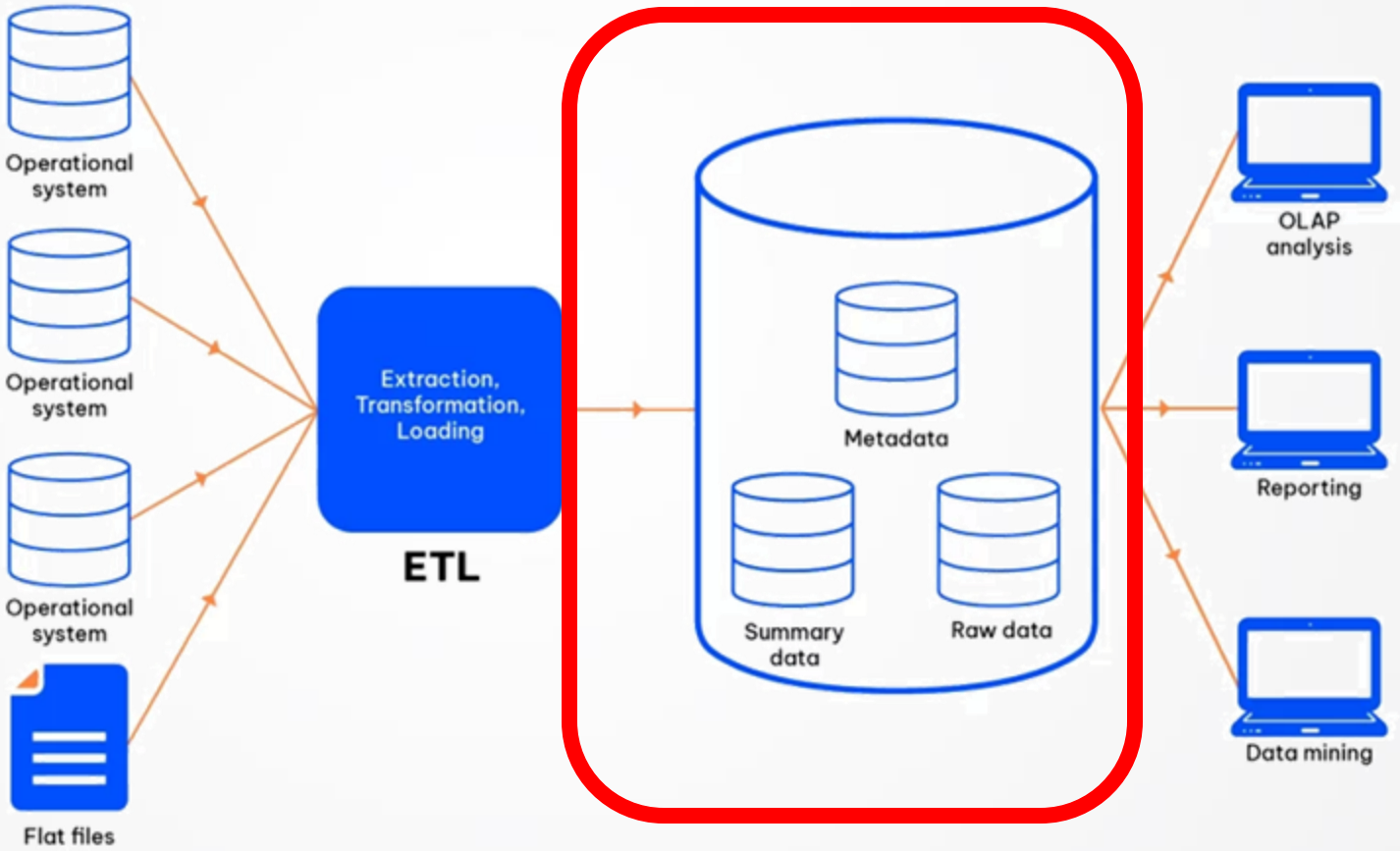
ETL process: Data extraction



ETL process: Data transformation



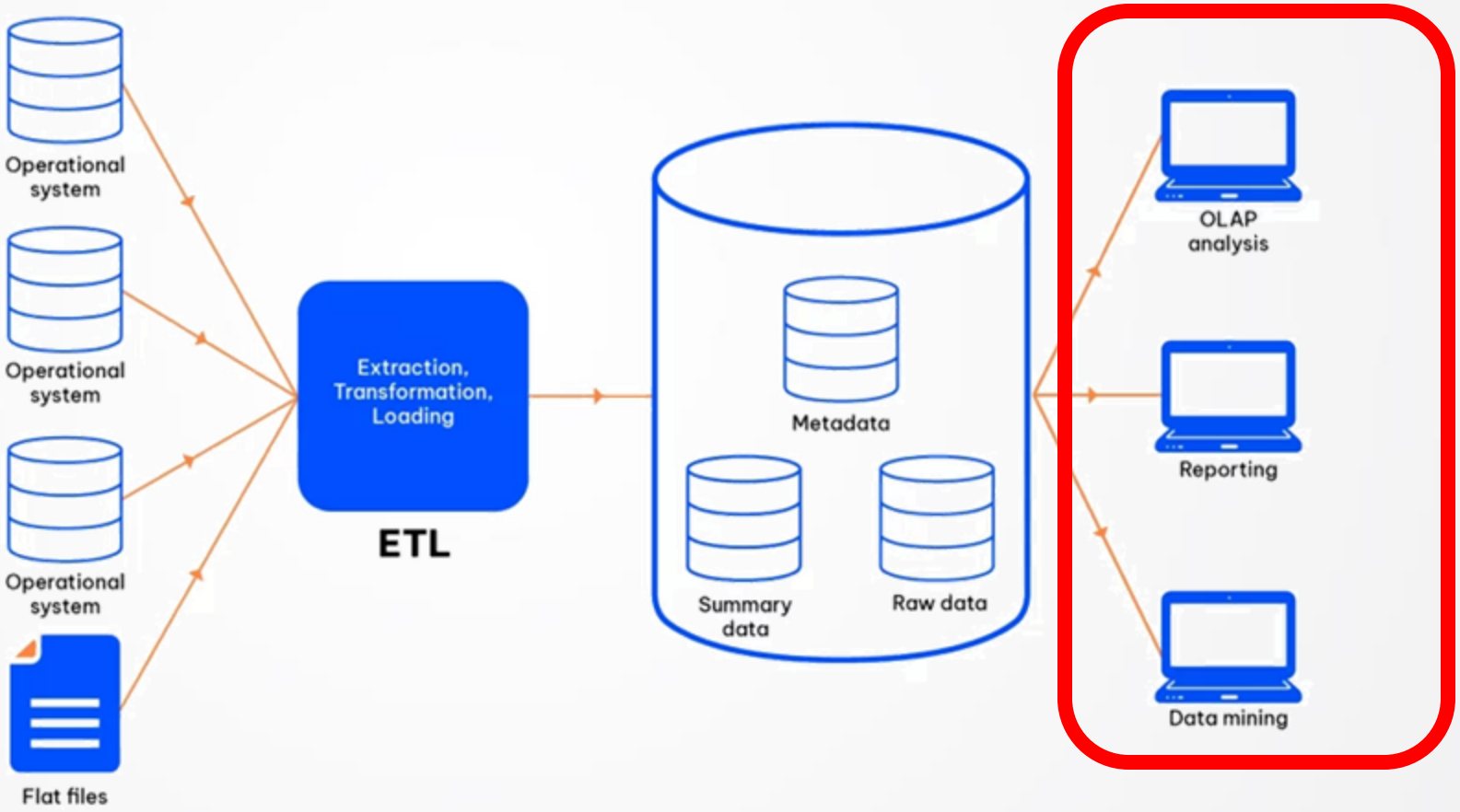
ETL process: Data Loading in Data warehouse





Data analysis & Data Visualization

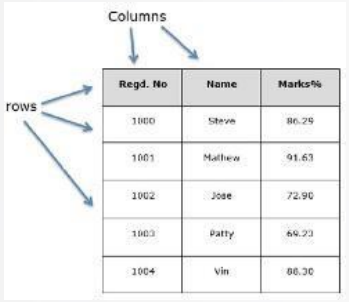
Data analysis & Data Visualization



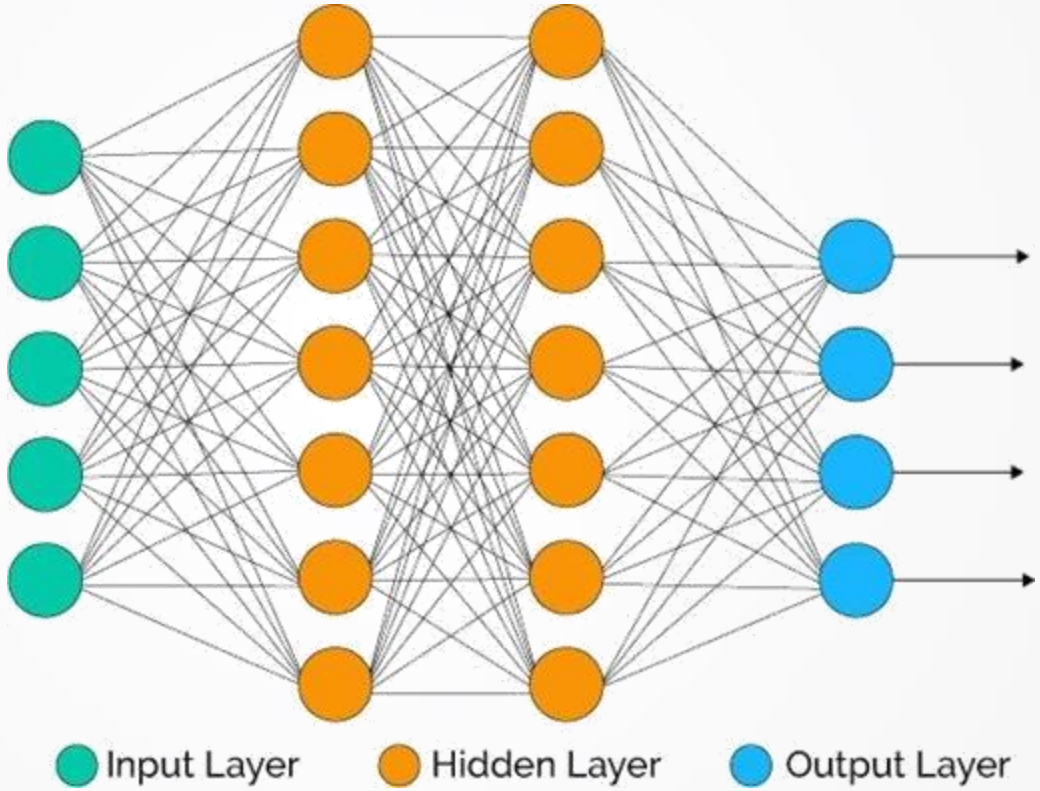
Prediction



Neural Network Model



Financial Dataset





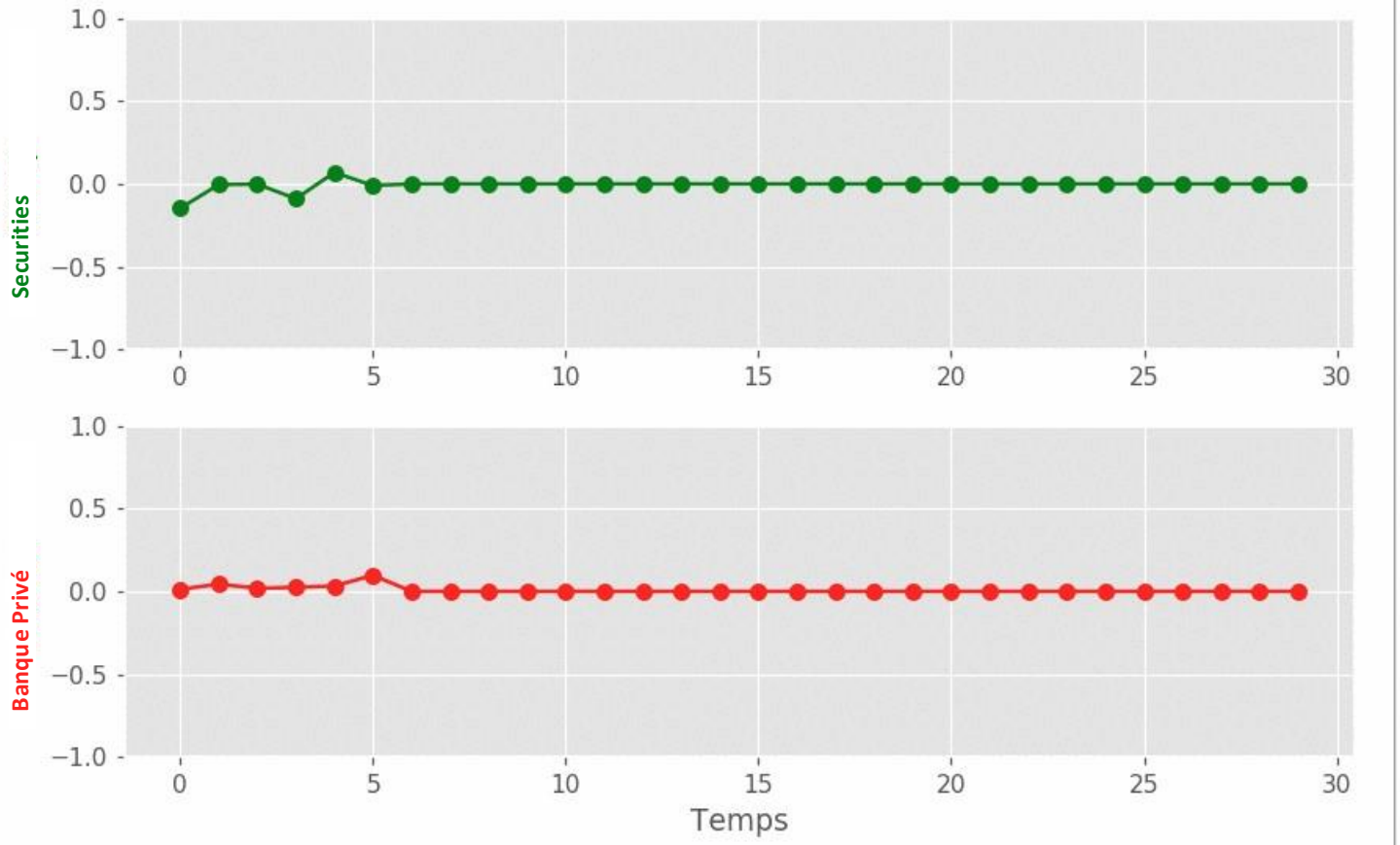
Client’s segment

Real-time sentimental analysis



Analyse de sentiment en temps réel sur Twitter

Visualisation



Paramètres

Mot-clé à chercher (1) : trump Securities

Mot-clé à chercher (2): Banque Privé

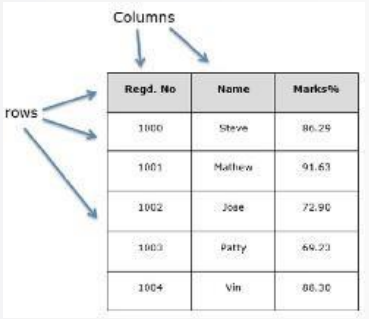
Lancer l'analyse !

	Tweet	Timestamp	Polarité	Subjectivité
1048155571923	@GuyFromV @Loreml	153873473679	0.3571	0.571428
1048155572086	RT @REALpunknews:	153873473683	0.25	0.333333
1048155572246	RT @SandraTXAS: "I s	153873473686	-0.035	0.301666
1048155574536	RT @AP: New York's a	153873473741	0.189(	0.535606
1048155574506	RT @suemitsu: TRUM	153873473740	0.0	0.0
1048155575157	RT @brutofficiel: Emr	153873473756	0.0	0.0
1048155575522	RT @SandraTXAS: Th	153873473765	-1.0	1.0
1048155575228	RT @Fahrenthold: The	153873473758	0.65	0.625
1048155575492	RT @suemitsu: TRUM	153873473764	0.0	0.0
1048155576025	RT @ChristieC733: Se	153873473777	0.25	0.75

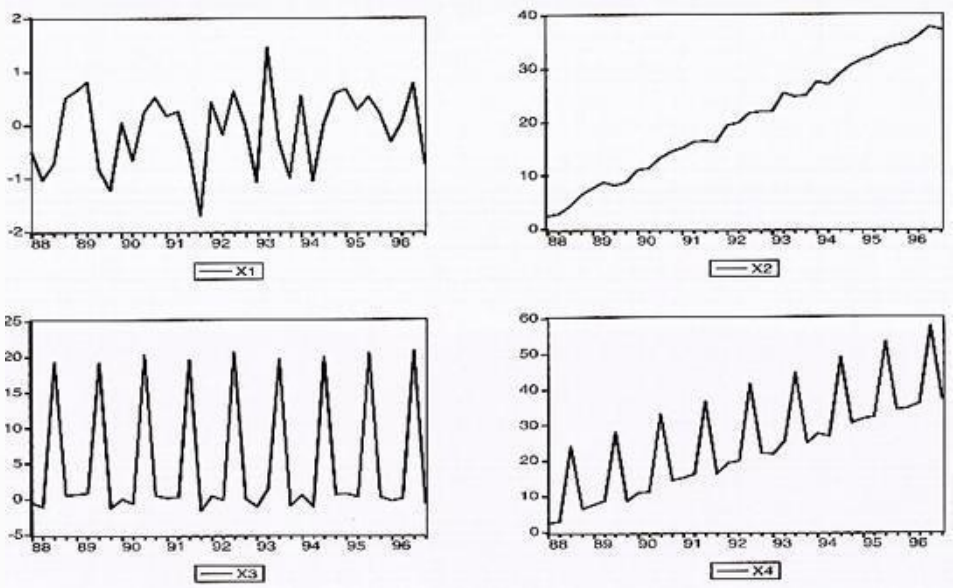
Quitter le programme

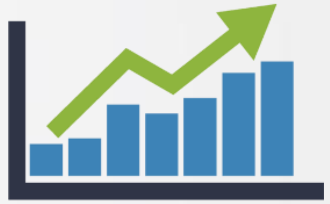
Prediction





Instruments Time Series





Forecasted Instruments

Scoring





Profil 1 : 100%



Profil 2 : 18%



Profil 3 : 92%



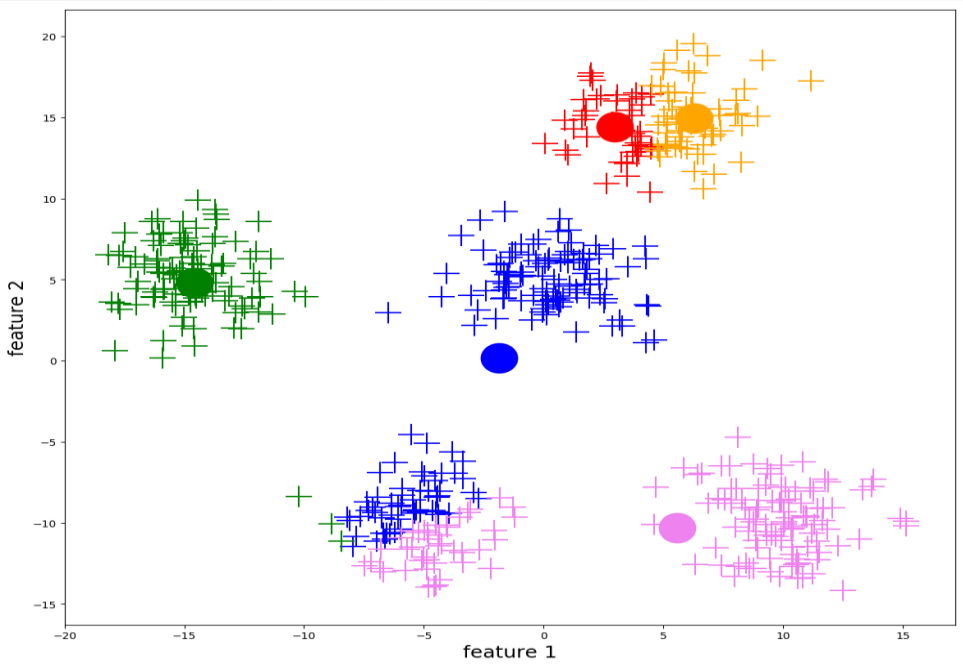
Profil 4 : 60%



Profil 5 : 87%

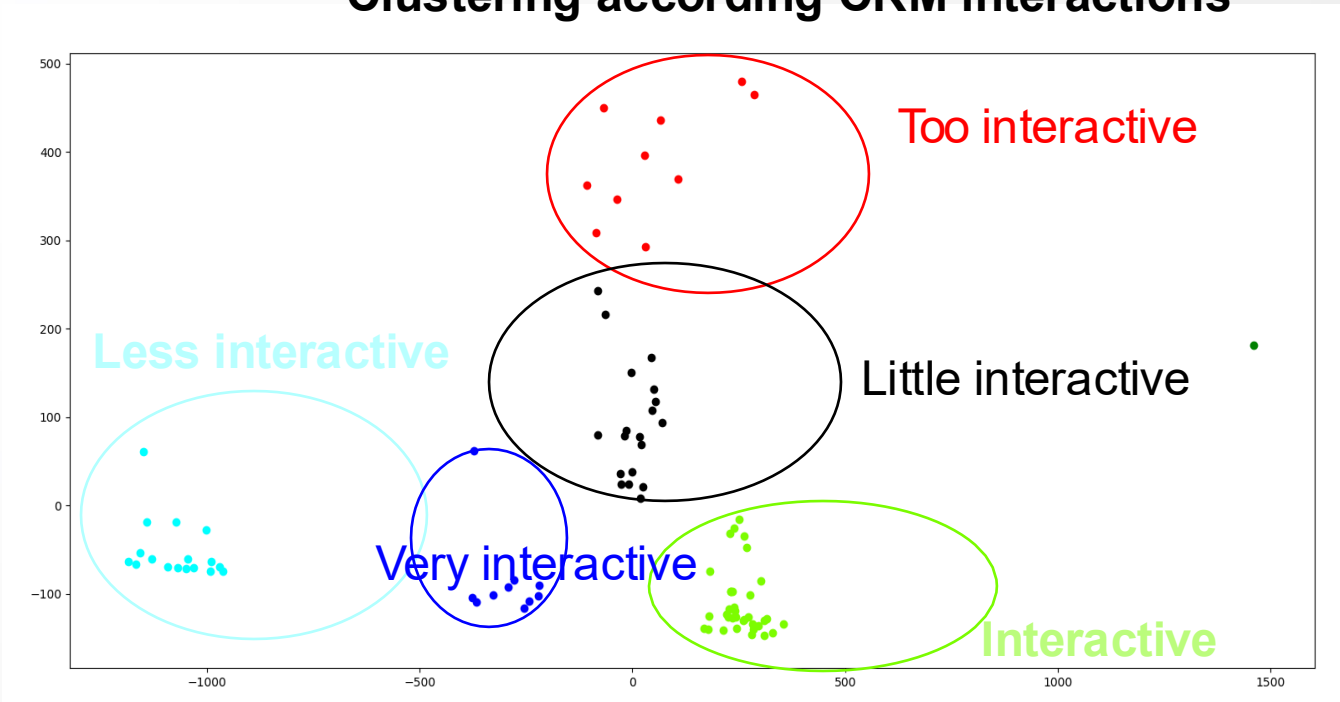
Clustering





Clustering according financial movements

Clustering according CRM interactions



Data Visualization



Customer satisfaction report

Metricalist | Customer Satisfaction Report

Get insights from your survey. Analyze your NPS, CSAT and Customer Sentiments.

Net Promoter Score

Promoters 45 (42%)

Passives 47 (44%)

Detractors 15 (14%)

3.85 Overall Rating

107 Survey Responde...

00:04:12 Avg Completi...

Customer Satisfaction Rating

Based on Total Response Received

Customer Sentiment

Positive Neutral Negative

Daily Sentiments

Daily NPS

5 Pending Surveys

Sent	Invitee	Status	Follow Up
21-06-2020	support@metricalist.com	Read	
21-06-2020	hello@metricalist.com	Started	
21-06-2020	admin@metricalist.com	Started	
21-06-2020	info@metricalist.com	Started	
23-06-2020	demo@metricalist.com	Read	

Feedback Cloud

Microsoft Power BI 1 of 5 99%

Data Visualization



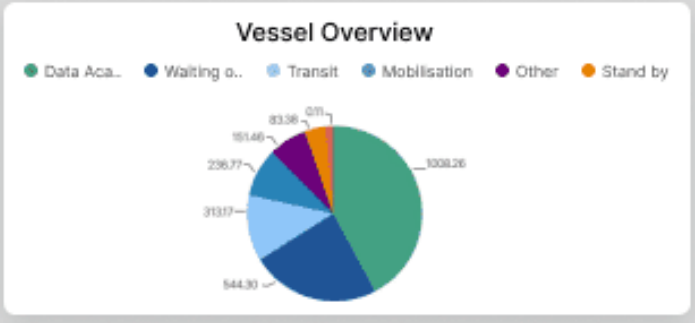
Shipping Industry Business Intelligence Dashboard

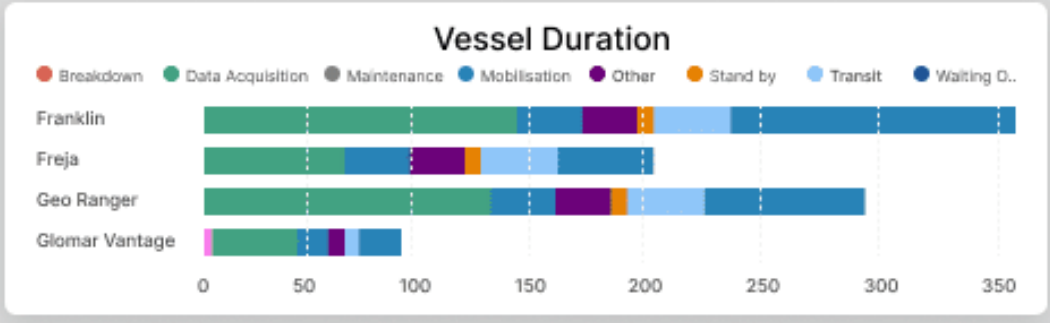
Vessel Detailed
Last Update : 2024-01-29 15:09

Projects	Projects Type	Vessel Type	Vessel	Year
All projects	Commerdal	All vessel	All vessel	2023

Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec

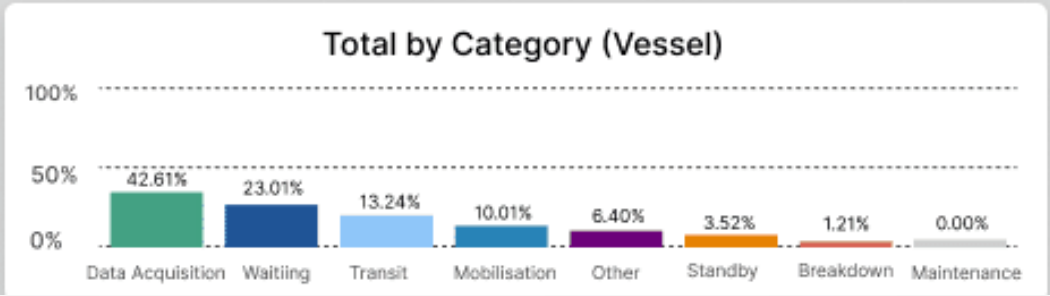
Maintenance (%)	Breakdown (%)	Data Acquisition (%)	Transit (%)	Mobilisation (%)
0.00%	1.21%	42.61%	13.24%	10.01%





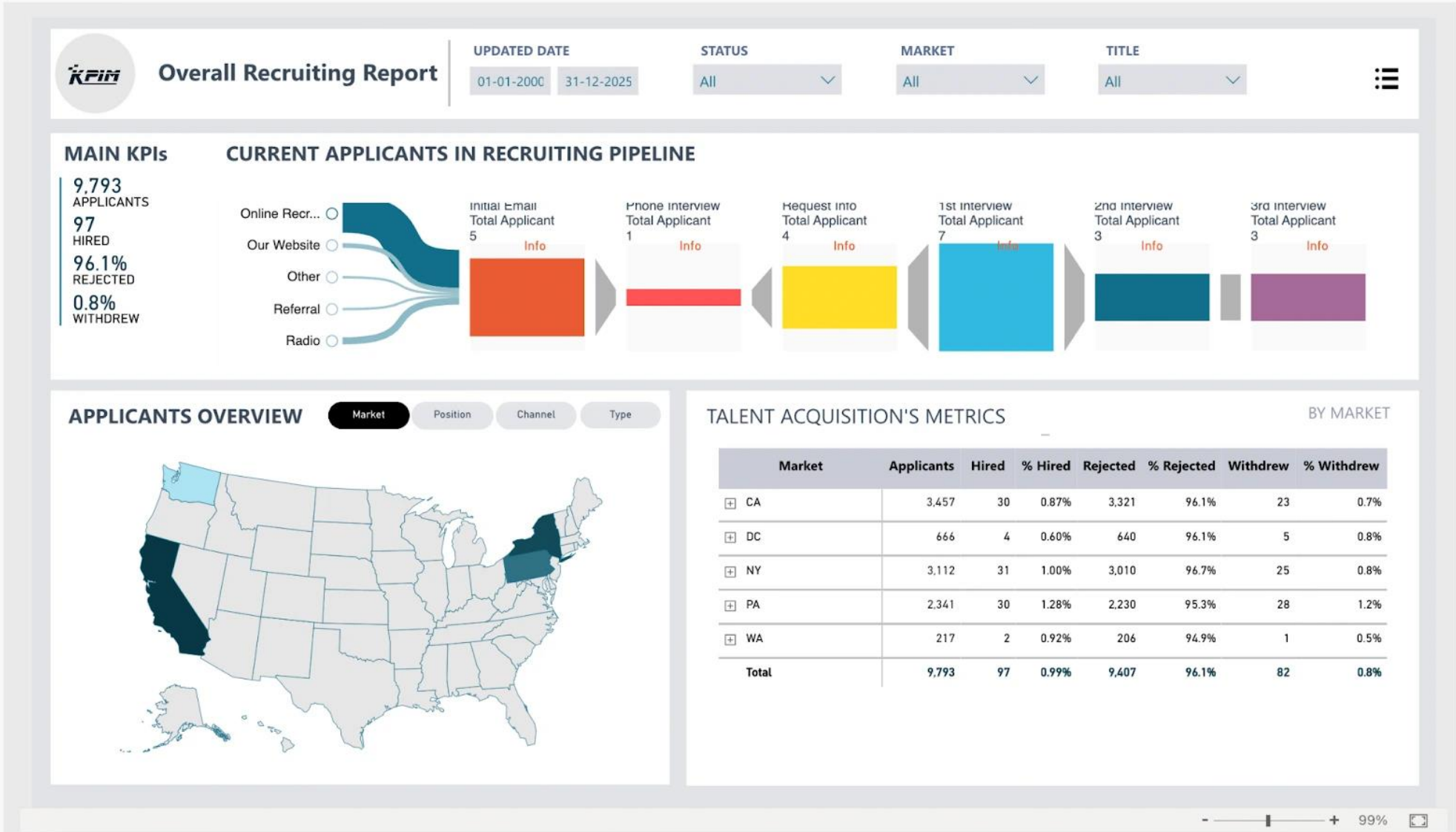
Vessel duration

Category	Vessel	Duration(in days)
2D UHRS Infill	Franklin	0.76
As-Found Survey	Normand Superior	57.15
Biological Sampling	Deep Helder	0.51
Boulder Removal	Normand Superior	0.83
Breakdown	African Spirit	1.27
Total		**2,365.99**



Data Visualization





The doashboard provides deep insight into the recruiting process with four main KPIs:

- Total applicants
- Total hired
- Total rejected
- Withdrawal during the process

Data Visualization







Web application

Mobile application

Tools











Evaluation

3 scores/Marks === 3 skills:

- Python for Data Engineering (2 credits)
  - The average of python evaluations
- Data Warehousing (3 credits)
  - The average of DWH evaluations
- Machine Learning basics (3 credits)
  - The average of ML evaluation.

** Multiple skills can be assessed within a single evaluation session

Projects

Project 1: Sport

- This project is based on a dataset covering padel tournament results, including player information, match scores, and team rankings.
- The goal is to analyze player and pair performance, identify playing trends, and extract meaningful statistics on tournament competitiveness.
- The data also opens the door to predictive modeling, such as estimating a pair's likelihood of winning a given match.

Project 2: health retailers

- In a context of digital transformation in the health and wellness sector, health retailers generate a significant volume of transactional data on a daily basis, stemming from sales, orders, inventory, products, suppliers, promotions, and, depending on the systems used, customer-related information.
- Leveraging this data represents a major opportunity to improve the commercial and operational performance of a drugstore.
- This project consists of designing a decision-support and analytics platform combining Business Intelligence (BI), Data Engineering, and Artificial Intelligence (AI) to centralize, transform, analyze, and add value to the transactional data of a drugstore located in France.

Project 3: NYC TLC Taxi Trips

- The New York City Taxi & Limousine Commission (TLC) publishes detailed, real-world trip records covering millions of rides across the city each month.
- Each record includes rich attributes such as pickup and drop-off timestamps and locations, trip distance, fare amount, payment type, and passenger count, offering a granular view of urban mobility patterns.
- The data is released monthly in Parquet format.
- This dataset offers strong potential for analyzing demand patterns, fare dynamics, and traffic trends across time and space, as well as for building predictive models such as fare estimation, trip duration forecasting, or demand hotspot detection.

Project 4: GitHub archive

- GitHub Archive is a public project that continuously records and archives the full public GitHub event stream, offering a detailed historical record of activity across the platform.`
- It captures over 15 distinct event types, including pushes, forks, issue creations, comments, pull requests, and membership changes, providing rich insight into how developers and organizations collaborate on open-source and public repositories.
- Data is released as hourly compressed JSON archives, with historical records available going back to 2011, enabling large-scale longitudinal analysis of developer behavior and project evolution.
- This dataset opens up strong opportunities for analyzing open-source community dynamics, contributor engagement trends, repository popularity over time, and collaboration patterns, as well as for building predictive models such as forecasting project activity spikes or identifying emerging influential repositories

Project 5: Brazilian E-commerce

- This dataset offers a comprehensive view of Brazilian e-commerce activity, drawing on approximately 100,000 real orders placed through the Olist Store across various Brazilian marketplaces between 2016 and 2018.
- The underlying data reflects genuine commercial transactions and has been anonymized for privacy purposes, with any mentions of companies or business partners in review text replaced by the names of great houses from Game of Thrones.
- A complementary Marketing Funnel dataset is also available separately, and the two can be merged to analyze orders through a marketing lens as well.

Project 6: Road safety

- This dataset provides detailed, long-term records of road collisions across Great Britain dating back to 1979, along with information on the vehicles and casualties involved in each incident.
- The statistics cover only personal injury collisions that occurred on public roads and were officially reported to and recorded by the police, following the STATS19 collision reporting system.
- All non-sensitive fields have been made publicly available, offering a rich foundation for analyzing accident trends, contributing factors, and road safety patterns over several decades.
- This dataset supports a wide range of analyses, from identifying high-risk locations and time periods to studying the relationship between casualty severity and factors such as vehicle type, weather conditions, or road characteristics, as well as building predictive models for accident risk.


ANY
QUESTIONS?